1.) Consider the function $F(x) = \sqrt{x^2+1}$. Which of the differentiation rules that we have learned so far (power, sum, difference, constant multiple, product, quotient) could we use to calculate $F'(x)$?

2.) We can think of F as a composition $F = f \circ g$, where $F(x) = f(g(x))$. What are f and g? To distinguish the domains of these two functions (recall that the domain of f must be the range of g in order for $f \circ g$ to be defined), write $f(u)$ and $g(x)$. Can you find $f'(u)$ and $g'(x)$?

Given a composition $F = f \circ g$ of differentiable functions f and g, it would be very useful to have a rule to compute the derivative of F in terms of the derivatives of f and g. We will now try to find such a rule using the following strategy: (i) First, consider compositions $F = f \circ g$ for which F' may be computed using the rules we already know. (ii) For each such F, relate $F'(x)$ to $f'(u)$ and $g'(x)$. (iii) Repeat until we find a pattern, from which we infer a rule. (iv) Prove that this rule holds for any composition $F = f \circ g$.

3.) Consider the function $F(x) = \sin(2x)$.

(a) If we view F as a composition $F = f \circ g$, what are f and g?

(b) Compute $f'(u)$ and $g'(x)$.

(c) Use the trig identities $\sin(2x) = 2\sin(x)\cos(x)$ and $\cos(2x) = \cos^2(x) - \sin^2(x)$ to find $F'(x)$.

(d) Relate $F'(x)$ to $f'(u)$ and $g'(x)$. (That is, given $f'(u)$ and $g'(x)$, is there a way to combine these to obtain $F'(x)$?)

♢ **Check your answers with your instructor before moving on.**

4.) Here is a more general example. Let $g(x)$ be a differentiable function.

(a) Use the product rule to compute $\frac{d}{dx}(g(x))^2$.

(b) Use the product rule and your answer to part (a) to compute $\frac{d}{dx}(g(x))^3$. (*Hint*: View $(g(x))^3$ as the product $(g(x))^3 = (g(x))^2 g(x)$.)

(c) Based on your answers to parts (a) and (b), what is $\frac{d}{dx}(g(x))^n$? **(Optional: Prove this by induction on n on a separate sheet of paper.)**

(d) View $F(x) = (g(x))^n$ as a composition of two functions f and g. What is $f(u)$?

(e) Relate $F'(x)$ to $f'(u)$ and $g'(x)$.

5.) Let $F = f \circ g$ be a composition of differentiable functions f and g. Make a conjecture for $F'(x) = (f \circ g)'(x)$. What are the differentiability requirements of f' and g' for $F'(x)$ to exist?

♢ **Check your answers with your instructor before moving on.**

Chain Rule: If g is differentiable at x and f is differentiable at $g(x)$, then the composite function $F = f \circ g$ defined by $F(x) = f(g(x))$ is differentiable at x and $F'(x)$ is given by the product

$$F'(x) = f'(g(x))g'(x).$$

6.) Compute $F'(x)$ if $F(x) = \sqrt{x^2+1}$.

7.) Differentiate (a) $y = \sin(x^2)$ and (b) $y = \sin^2(x)$.

8.) Differentiate $y = (x^3-1)^{100}$.

♢ **Check your answers with your instructor before moving on.**

9.) Find $f'(x)$ if $f(x) = \dfrac{1}{\sqrt[3]{x^2+x+1}}$.

10.) Find the derivative of the function $g(t) = \left(\dfrac{t-2}{2t+1}\right)^9$.

11.) Differentiate $y = (2x+1)^5(x^3-x+1)^4$.